

TrES-3b

R-1943

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_140423 · created 2026-08-06T14:50:45+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2456770.8531 +/- 0.0011 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.089 +/- 0.049
Transit depth (Rp/Rs)^2	0.79 +/- 0.88 [%]
Orbital Inclination (inc)	86.96 +/- 2.96
Airmass coefficient 1 (a1)	1.029 +/- 0.026
Airmass coefficient 2 (a2)	-0.018 +/- 0.019
Scatter in the residuals of the lightcurve fit is	2.53 %
Best Comparison Star	#3 - [452, 124]
Optimal Aperture	2.85
Optimal Annulus	8.81
Transit Duration (day)	0.071 +/- 0.013

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.089 ± 0.049 vs 0.16309 (deviation 1.51σ)
Duration fit vs published	0.071 d vs 0.0567 d (deviation 1.1σ)
Mid-time O–C	-1.7 min
Depth SNR	1.8
Residual scatter	2.53 %

Light curve

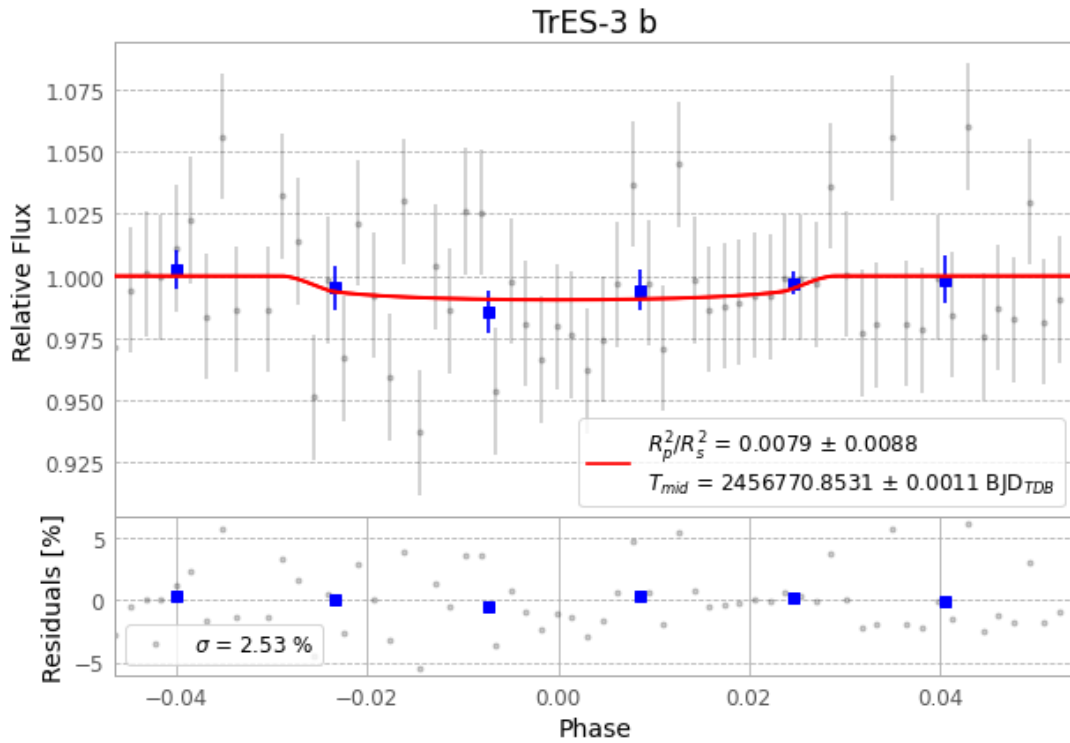


Figure 1 — FinalLightCurve_TrES-3 b_2014-04-22.png (SHA-256 228b0b63c3fa51b6...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [246, 232], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 7403b9ea22e1690d...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

64 FITS frames (42 MB), TRES-3140423065712.FITS → TRES-3140423100612.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-140423.log (587a46caae90...), FinalLightCurve_TrES-3 b_2014-04-22.png (228b0b63c3fa...), FinalLightCurve_TrES-3 b_2014-04-22.csv (619b6545e461...)

Compute resources

Wall time 136.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-06 14:50Z.